EXT GROUPS BEFORE AND AFTER REDUCTION MODULO THE POWERS OF AN IDEAL ARE PRO-ISOMORPHIC

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement span includes the author's own \begin{reference} block recording the email correspondence in which the result was found; that block is delivered verbatim and produces no unresolved citation. The author writes '(precise formulation omitted)' and 'A straightforward diagram chase (omitted) shows that the kernel and cokernel of the dotted arrow are annihilated buy I^c'; both are kept verbatim, including the typo 'buy' for 'by'. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The category C of systems with the comparison lemma for maps whose kernel and cokernel are annihilated by a fixed power of I. the target category at statement lines 29208-29209 and invoked at proof lines 29228-29244. Is the category in which the isomorphism is asserted and the tool that promotes bounded-annihilation maps to isomorphisms.
- The Artin-Rees consequence for the homology of a complex, giving the constant c and the almost inverse maps. invoked at proof lines 29224-29225. Produces the constant c that makes the systems (K(n)) and (K_n) isomorphic.
- Mittag-Leffler comparison for the Hom systems before and after reduction. invoked at proof lines 29277-29280. Bounds the kernel and cokernel of the two Hom comparison maps by I^c-torsion uniformly in n.

Source reference. Original lemma 28867: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28867–28867. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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Source reference. Original lemma 29060: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29060–29060. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 29168: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29168–29168. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 29141: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29141–29141. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let I be an ideal of the Noetherian ring A. Let M and N be finite A-modules. Write $A_n = A/I^n$, $M_n = M/I^nM$, and $N_n = N/I^nN$. For every $i \geq 0$ the objects*

$$\{\mathrm{Ext}^i_A(M,N)/I^n \mathrm{Ext}^i_A(M,N)\}_{n\geq 1} \quad \textit{and} \quad \{\mathrm{Ext}^i_{A_n}(M_n,N_n)\}_{n\geq 1}$$

are isomorphic in the category $\mathcal{C}$ of Remark 29141.

Proof. Choose a short exact sequence

$$0 \to K \to A^{\oplus r} \to M \to 0$$

and set $K_n = K/I^nK$. For $n \geq 1$ define $K(n) = \mathrm{Ker}(A_n^{\oplus r} \to M_n)$ so that we have exact sequences

$$0 \to K(n) \to A_n^{\oplus r} \to M_n \to 0$$

and surjections $K_n \to K(n)$. In fact, by Lemma 28867 there is a $c \geq 0$ and maps $K(n) \to K_n/I^{n-c}K_n$ which are "almost inverse". Since $I^{n-c}K_n \subset K_n[I^c]$ these maps which witness the fact that the systems $\{K(n)\}_{n\geq 1}$ and $\{K_n\}_{n\geq 1}$ are isomorphic in $\mathcal{C}$.

We claim the systems

$$\{\mathrm{Ext}^i_{A_n}(K(n),N_n)\}_{n\geq 1} \quad \text{and} \quad \{\mathrm{Ext}^i_{A_n}(K_n,N_n)\}_{n\geq 1}$$

are isomorphic in the category $\mathcal{C}$. Namely, the surjective maps $K_n \to K(n)$ have kernels annihilated by I^c and therefore determine maps

$$\mathrm{Ext}^i_{A_n}(K(n),N_n) \to \mathrm{Ext}^i_{A_n}(K_n,N_n)$$

whose kernel and cokernel are annihilated by I^c. Hence the claim by Lemma 29168.

For $i \geq 2$ we have isomorphisms

$$\mathrm{Ext}^{i-1}_A(K,N) = \mathrm{Ext}^i_A(M,N) \quad \text{and} \quad \mathrm{Ext}^{i-1}_{A_n}(K(n),N_n) = \mathrm{Ext}^i_{A_n}(M_n,N_n)$$

In this way we see that it suffices to prove the lemma for $i = 0, 1$.

For $i = 0, 1$ we consider the commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathrm{Hom}(M,N) & \longrightarrow & N^{\oplus r} & \xrightarrow{\varphi} & \mathrm{Hom}(K,N) & \longrightarrow & \mathrm{Ext}^1(M,N) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & & & & & \mathrm{Hom}(K_n,N_n) & & \\
 & & & & & & \uparrow & & \\
0 & \longrightarrow & \mathrm{Hom}(M_n,N_n) & \longrightarrow & N_n^{\oplus r} & \longrightarrow & \mathrm{Hom}(K(n),N_n) & \longrightarrow & \mathrm{Ext}^1(M_n,N_n) \longrightarrow 0
\end{array}$$

By Lemma 29060 we see that the kernel and cokernel of $\mathrm{Hom}(M,N)/I^n\,\mathrm{Hom}(M,N) \to \mathrm{Hom}(M_n,N_n)$ and $\mathrm{Hom}(K,N)/I^n\,\mathrm{Hom}(K,N) \to \mathrm{Hom}(K_n,N_n)$ and are I^c-torsion for some $c \geq 0$ independent of n. Above we have seen the cokernel of the injective maps $\mathrm{Hom}(K(n),N_n) \to \mathrm{Hom}(K_n,N_n)$ are annihilated by I^c after possibly increasing c. For such a c we obtain maps $\delta_n : I^c\,\mathrm{Hom}(K,N)/I^n\,\mathrm{Hom}(K,N) \to \mathrm{Hom}(K(n),N_n)$ fitting into the diagram (precise formulation omitted). The kernel and cokernel of δ_n are annihilated by I^c after possibly increasing c since we know that the same thing is true for $\mathrm{Hom}(K,N)/I^n\,\mathrm{Hom}(K,N) \to \mathrm{Hom}(K_n,N_n)$ and $\mathrm{Hom}(K(n),N_n) \to \mathrm{Hom}(K_n,N_n)$. Then we can use commutativity of the solid diagram

$$\begin{array}{ccccccc}
\varphi^{-1}(I^c\,\mathrm{Hom}(K,N)) & \xrightarrow{\varphi} & I^c\,\mathrm{Hom}(K,N)/I^n\,\mathrm{Hom}(K,N) & \longrightarrow & I^c\,\mathrm{Ext}^1(M,N)/I^n\,\mathrm{Ext}^1(M,N) & \longrightarrow & 0 \\
\downarrow & & \downarrow{\scriptstyle \delta_n} & & \vdots & & \\
N_n^{\oplus r} & \longrightarrow & \mathrm{Hom}(K(n),N_n) & \longrightarrow & \mathrm{Ext}^1(M_n,N_n) & \longrightarrow & 0
\end{array}$$

to define the dotted arrow. A straightforward diagram chase (omitted) shows that the kernel and cokernel of the dotted arrow are annihilated buy I^c after possibly increasing c one final time. □